Whether Chemical Warfare Agent Simulant Designations Look Supported Depends on Which Compounds Are in the Comparison

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Abstract

Chemical warfare agent detection is developed almost entirely on simulants: compounds accepted as agent-like and safe enough to handle in an ordinary laboratory. We tested whether that acceptance has a measurable basis. For eleven authentic agents we computed structural and GFN2-xTB electronic descriptors and asked, by exact permutation over every possible designation, whether three incumbent simulants sit closer to the agents than compounds drawn at random. On our primary thirty-five-compound panel, which includes five further compounds the sensor literature also uses as simulants, no agent has a valid match at any threshold and no test falls below $p = 0.15$. Resampling the descriptors gives the incumbent three a median of zero matched agents against five for the eight the field uses, the difference positive in 1000 of 1000 replicates. The distances barely move between panels; what changes is that something closer is present. A twenty-eight-compound panel omitting those five yields three matches and a smallest nominal p of 0.025, surviving one of five defensible multiplicity schemes. Two of the three were designated for measurements that are not detection, and the third was not designated at all. A defensible claim must name the originating measurement and fix the panel, space and statistic in advance.

Keywords: chemical warfare agent simulant, descriptor space, DMMP, exact permutation test, multiplicity, acute exposure guideline level, dual-use

1. Introduction

Work on detecting chemical warfare agents (CWAs) is rarely done with CWAs. Outside a small number of licensed facilities, sensor arrays, sorbents and sampling protocols are developed, optimised and published using simulants: compounds chosen to behave like an agent while being safe enough for an ordinary laboratory. The choice of simulant therefore sets a ceiling on what a published detection result can be taken to mean, and that choice is made by a convention rather than by a measurement.

The dependence is structural, not incidental. The standing reviews of agent destruction and detection organise a large literature in which reactivity and sensing chemistry are established on surrogates and extrapolated to agents [1, 2], and the same pattern runs through the chromogenic,

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fluorogenic and supramolecular recognition chemistry developed for nerve agents [3, 4, 5]. Much current field instrumentation descends from a colorimetric sensor-array lineage established on ordinary analytes and odours [6, 7, 8] and then transferred to the agent problem, so the compounds on which an array is characterised are inherited from that lineage rather than fixed by the chemistry. Our own work sits on both sides, with arrays developed on surrogates [9, 10] and optical detection closed against authentic Novichok compounds [11]; the second is rare enough that the first is where most of the field's evidence is generated.

What a simulant is for differs by subfield, and so does what makes one adequate. For environmental fate and transport the criterion has been reviewed explicitly [12]; for adsorption and catalytic hydrolysis in metal–organic frameworks it is the simulant that establishes the mechanism and the kinetics [13, 14]; in decontamination the organophosphorus simulant dimethyl methylphosphonate (DMMP) is the compound whose removal is quantified [15]. In every case the inference runs from the simulant to the agent, and the warrant for it is a resemblance argument.

The convention is property matching. A recent critical review of liquid, low-toxicity simulants states the criterion plainly: align physicochemical properties such as hydrophobicity, volatility and solubility with those of the agent, and the simulants it assesses are described as presenting minimal toxicity risks [16]. That review is confined to liquid-phase sampling, but the same criterion underwrites simulant choice in vapour-phase detection, where the quantity of interest is not partitioning but a chemical response.

Two assumptions are built into this practice, and both are testable. The first is that resemblance in structural and physicochemical descriptors implies resemblance in the property a sensor actually reports. For adsorption in metal–organic frameworks this has been examined directly: simulating four common simulants and two nerve agents across 2969 frameworks showed that simulants reproduce agent adsorption behaviour only partially, and that the agreement depends on which framework and which agent [17]. The analogous question for the electronic properties that govern a chemical response has not been asked. The second assumption is that a simulant is safe, and it is usually supported by the absence of an agent-class hazard label rather than by a toxicity value, although where simulants are handled at scale the protective equipment is specified against measured permeation and exposure data [18].

Both assumptions can be checked against measurements that already exist. The electronic structure of small molecules is accessible at semi-empirical tight-binding cost [19, 20, 21], and acute inhalation toxicity has been derived by a formal regulatory process for several hundred chemicals, including the agents themselves [22]. Ranking compounds along a sparse regulator-derived hazard axis bounded by an explicit applicability domain is the approach we have applied elsewhere [23]; here it is turned on the compounds the detection field uses on itself. We ask where simulants and agents actually sit relative to one another, how much of that answer is fixed by the data rather than by the analyst's choice of compound panel, descriptor space and statistic, whether the incumbent simulants are in fact safe and whether that claim rests on a value or on an absence, and whether fidelity and safety are coupled.

The first question turned out to have a different shape than we expected. We began by comparing descriptor spaces, found a difference, and then found on audit that it did not survive extending the compound panel, correcting for the number of statistics computed, or removing a descriptor that was an exact function of two others in the same space. What the data support is reported below as such, with the earlier reading alongside it, because the way a nominal result of this kind arises is part of what a reader of this literature needs.

Dual-use statement.. Throughout, the optimisation direction is fixed: fidelity to the detection-relevant properties is to be maximised while phosphorus electrophilicity, acetylcholinesterase reactivity, alkylating capacity and predicted mammalian toxicity are to be minimised. No analysis here ranks compounds by potency. The structural filter in Section 2.7 excludes candidates and never orders them, and we did not extend the toxicity model of Supplementary Section S0.2 to the organophosphorus class, because a model predicting mammalian lethality for structures not yet made would invert the direction of the work. That a predictive model trained for safety can be run in reverse is not hypothetical [24], and how much of such a method to disclose is itself a live problem [25]. Mechanistic modelling of organophosphorus lethality is established and separately published, including by the present authors [26]; what is withheld is its use as a design objective over novel structures, not the literature's existence.

2. Methods

2.1. Compound set and the two panels

Eleven authentic agents are common to both panels: sarin GB, soman GD, tabun GA, cyclosarin GF, VX, sulfur mustard HD, lewisite 1 L, hydrogen cyanide AC, cyanogen chloride CK, phosgene CG and chloropicrin PS. Three compounds are treated as the incumbent simulants: DMMP, dichloromethane and dipropylene glycol monomethyl ether (DPGME).

We call these three incumbent rather than designated, because the documentary record is more specific than either word suggests. Searching for an instrument that designates them as chemical warfare agent simulants, we found that two of the three carry a documented designation for a purpose other than the one they are now used for, and the third carries none.

DMMP was identified as a test simulant for canister and cartridge filtration against GB and HD in a NIOSH respirator programme [27], a document that also states the simulant information it provides is not to be used as a recommendation of any simulant, and whose property at issue is adsorptive breakthrough rather than detection chemistry. DPGME appears as "a preferred G agent simulant" in a patent assigned to the United States Department of the Army [28], in a training composition formulated with a fluorescent dye and justified on ion-mobility detectability, so the application is field training rather than sensor validation. For dichloromethane we found no designating document of any kind: it is absent from the NIOSH candidate list and from the Man-in-Simulant Test programme, and present in this role, as far as we can establish, only in the research literature.

We state what that search covered, because a negative claim about documents is only as good as its scope. We searched the standard-setting and test-procedure literature reachable without a subscription or a clearance: NIOSH and NPPTL respirator programme documents, US Army and DEVCOM technical reports indexed by the Defense Technical Information Center, United States patents assigned to the Department of the Army, National Academies assessments of Department of Defense test programmes, ASTM and NATO standards catalogues, and the peer-reviewed simulant literature cited here. Two Test Operations Procedures whose titles bear on the question (TOP 08-2-196A and TOP 10-2-022B) we could not obtain, and national CBRN defence manuals are generally not public. A designation in one of those would change the paragraphs above but not the measurements below.

Two observations follow, and they shape how the rest of this paper should be read. A designation is specific to a measurement: a compound chosen because it breaks through a carbon bed like sarin has been validated for adsorption, and nothing in that transfers to an optical or

electrochemical response, while a compound chosen because it lights up an ion-mobility detector in a training lane has been validated for that detector and not for a colorimetric array. The sensor literature using these three is using designations issued for other purposes. The compounds with the clearest designations, in turn, are not these three: methyl salicylate is the simulant of the United States Army's Man-in-Simulant Test programme, chosen for sulfur mustard on 1994 penetration trials [29], and the NIOSH candidate list consists of DFP, DIMP, DEEP, TEP, dibutyl sulfide, CEES, malathion and methyl salicylate. Those compounds are in the second panel described below, not the first. A recent survey reaches the general form of the same conclusion, that simulants have been selected largely on their historical use and that no standard for chemical agent simulants exists [30].

The two panels differ only in the non-agent pool, and the difference is the point. The twenty-eight-compound panel adds fourteen interferents and solvents routinely reported in sensor-array selectivity panels. The thirty-five-compound panel adds, to those, five further compounds used as simulants in the sensor, sorbent and decontamination literature (diisopropyl fluorophosphate DFP, diethyl chlorophosphate DCP, diethyl cyanophosphonate DCNP, 2-chloroethyl ethyl sulfide CEES and methyl salicylate), classified as simulants rather than as interferents, together with two organophosphorus pesticides, dichlorvos and malathion. We class the pesticides as interferents because the literature we surveyed does not use them to stand in for an agent, and flag the decision as contestable: both carry the phosphoryl motif that the structural filter of Section 2.7 treats as agent-class. Section 3.1 reports how sensitive the result is to exactly this kind of decision.

The thirty-five-compound panel is the primary one, and the five it adds are the ones with documentary standing: DFP and methyl salicylate appear in the NIOSH candidate list quoted above, methyl salicylate is the Man-in-Simulant Test simulant [29], and all five are in current use. A comparison asking whether three compounds are close to the agents while excluding the obvious alternatives is not a test of those three. We report the twenty-eight-compound panel throughout with equal numerical detail, because what it shows beside the larger one is the paper's result, and not to support a conclusion. All class labels were read from a single canonical table and never assigned by hand.

2.2. Descriptors

Structural and physicochemical descriptors (51 after excluding run metadata) were computed with RDKit [31] from canonical SMILES, and include the Crippen atomic-contribution estimates of logP and molar refractivity [32], the fragment-based topological polar surface area [33] and the κ shape indices [34]. Interpretable descriptors were used throughout in preference to a substructural fingerprint such as ECFP [35], because the quantity of interest here is the position of a compound along named physical axes rather than a similarity score. Identifiers, formulae and molecular weights were resolved against PubChem [36]; the EPA CompTox Dashboard [37] was the intended second route but its programmatic interface was not reachable from either network used here, so identities rest on PubChem and on the identifiers printed in the regulatory source documents, and no property value was taken from a prediction service.

Electronic descriptors were computed with GFN2-xTB [20], the second-generation parametrisation of the geometry–frequency–noncovalent tight-binding family [19, 21], in the gas phase; the implicit solvation model available for these methods [38] was not applied, since vapour-phase response is the property of interest. Conformers were generated with ETKDGv3 [39, 40], optimised with MMFF94 [41] (UFF [42] for lewisite 1, the only compound lacking MMFF parameters), then optimised and evaluated at GFN2 level. Ensembles were enumerated from the distance-geometry stage rather than through a dedicated sampling workflow [43], as these

descriptors are not strongly conformation dependent. Properties for the 28-compound set were Boltzmann averaged over up to five conformers at 298.15 K with tight optimisation; the larger AEGL screening set used a single conformer with `-opt normal`. On the 28 compounds the two protocols agree to Pearson $r \geq 0.9857$ across the four electronic descriptors, with mean absolute differences of 0.029 to 0.052 eV for the orbital quantities and 0.11 D for the dipole, the largest single dipole discrepancy being 0.95 D at tabun. Which protocol an analysis uses is fixed by what it compares. The descriptor-space comparison of Section 3.1, which stays inside the 28 compounds, uses the Boltzmann-averaged values. The decoupling analysis of Supplementary Section S0.11 places the 28 and the AEGL pool in one space, so it uses the single-conformer values for all of them, including the 28; mixing protocols within a distance calculation would put part of the systematic difference above into the distances.

Four electronic descriptors are available for every compound: E_{HOMO}, E_{LUMO}, the global electrophilicity index $\omega = \mu^2/2\eta$ in the definition of Parr et al. [44], and the dipole moment. Four further descriptors exist only for the six phosphorus-bearing compounds: the Mulliken charge at phosphorus q_{P}, the charge on the phosphoryl oxygen, the P=O bond order in the Wiberg [45] and Mayer [46] sense, and the condensed Fukui index $f^{+}(\mathrm{P})$, which measures the susceptibility of that centre to nucleophilic attack [47, 48] and whose standing as a reactivity descriptor is reviewed elsewhere [49, 50]. These four are reported but excluded from any analysis spanning all 28 compounds.

2.3. Descriptor spaces and distances

Three spaces were compared: structural only, electronic only, and merged.

The electronic space carries three coordinates, not the four that frontier-orbital descriptor sets of this kind usually report. The fourth, the Parr electrophilicity index $\omega = \mu^2/2\eta$ with $\mu = (E_{\mathrm{HOMO}} + E_{\mathrm{LUMO}})/2$ and $\eta = (E_{\mathrm{LUMO}} - E_{\mathrm{HOMO}})/2$ [44], is an exact function of the two orbital energies that accompany it, so it adds no coordinate. We confirmed this numerically rather than by inspection: across the panel, $|\omega - \mu^2/2\eta|$ reaches at most 3.6×10^{-15} eV, which is floating-point noise. The correlation filter described below does not remove it, because $r(E_{\mathrm{LUMO}}, \omega) = -0.899$ falls below the $|r| > 0.95$ threshold; functional dependence and linear correlation are not the same condition, and a filter built on the second will pass a variable determined exactly by the first. Three principal components of the standardised four-variable set account for 97.6% of its variance, and removing ω leaves the rank order of pairwise distances almost unchanged (Spearman 0.932). We therefore take E_{HOMO}, E_{LUMO} and the dipole moment as the electronic space, and report ω and the condensed Fukui index as derived quantities in Supplementary Table S1 without admitting them to any distance. The four reactivity descriptors localised at phosphorus are undefined for the twenty-two panel compounds that contain none, so they cannot enter a distance computed over the whole panel either.

Within each space, descriptors with zero variance were removed, then one of each pair correlated at $|r| > 0.95$ was removed with preference for retaining a protected set (molar refractivity, logP, TPSA, hydrogen-bond acceptor count, four linear-solvation-energy surrogates and the electronic descriptors). The four surrogates are linear rescalings of quantities already in the set, namely the hydrogen-bond donor and acceptor counts, molar refractivity, and the sum of topological polar surface area and molar refractivity; they are not Abraham solvation parameters, and three of the four are removed by the correlation filter at $|r| = 1.00$ in every space. Remaining descriptors were standardised and distances taken as Euclidean. Two shape indices, κ_2 and κ_3, go negative for the smallest molecules (water, $\kappa_3 = -104.0$); repeating every analysis without them returns the permutation values of Section 2.4 to three decimal places (structural 0.028, electronic 0.275,

merged 0.025, on 36 and 39 descriptors instead of 38 and 41), so no result below rests on the pathological indices.

2.4. Nearest-neighbour comparison and permutation test

For each authentic agent we recorded the distance to the nearest incumbent simulant and to the nearest interferent. Because there are fourteen interferents and three simulants, the minimum of fourteen draws is smaller than the minimum of three draws by construction, and a paired test of the two distances is confounded by group size. We therefore enumerated all ways of designating three of the non-agents as simulants, computed the same statistic for each, and report the rank of the actual designation as an exact p-value. That is $\binom{24}{3} = 2024$ designations on the primary panel and $\binom{17}{3} = 680$ on the twenty-eight-compound panel. The null expectation for "an interferent is closer" follows from the group sizes alone and is 9.63 of 11 on the primary panel and 9.06 on the smaller one.

The hypothesis under test is directional, since property matching is claimed to place simulants closer to the agents than arbitrary compounds would be. The reported p is therefore one-sided, the fraction of designations whose mean d(nearest simulant) − d(nearest interferent) is at least as negative as the actual one's. The choice is consequential and we give all three conventions: scored two-sided as distance from the null mean in either direction the same statistic gives 0.029, 0.482 and 0.025 for the structural, electronic and merged spaces, and at twice the smaller tail it gives 0.056, 0.550 and 0.050, so whether the structural value sits inside or outside a conventional threshold depends on a choice a reader may make differently. This is the first of the analyst choices catalogued in Section 3.1 and the only one with a defensible default, the hypothesis as its proponents write it being directional.

2.5. Match criterion

Nearest-neighbour rank alone is not sufficient: in a 38-dimensional space a "nearest" simulant at a distance of 11.8 is not a match. We therefore normalised each distance by the median nearest-neighbour distance of the same space and define a valid match as a normalised distance ≤ 2 together with a distance smaller than that to every compound outside the designated set. The two conditions are not equally fragile: the threshold is a convention, the second condition is not. On the primary panel the incumbent simulants fail the second condition for all eleven agents, so the count there is zero at every threshold, including an unbounded one. We report the count at 1.5, 1.75, 2.0, 2.5 and 3.0 in Section 3.3, including the one setting at which it changes.

2.6. Descriptor bootstrap

The conclusions of Section 3.1 turn on differences that are small in absolute terms, 0.12 typical nearest-neighbour separations between DMMP and diisopropyl fluorophosphate at sarin, and a reader is entitled to ask how much of that the descriptor choice carries. We resampled the 41 merged-space descriptors with replacement 1000 times (seed `0xC0DE`), restandardised, recomputed the distance matrix and the median nearest-neighbour separation within each replicate, and recounted the matched agents. Resampling features rather than compounds is the relevant perturbation, because the compound set is fixed by the question and the descriptor set is not. The same replicates serve both simulant definitions, so the difference is paired.

2.7. Structural filter

A SMARTS filter was written to exclude candidates carrying agent-class motifs, following the structural-alert tradition in which a substructure associated with a mechanism of toxicity serves as a categorical screen [51, 52]; the motifs here are agent-class rather than generic and the filter excludes only. Tier A covers a phosphoryl group bearing a good leaving group (fluoride, cyanide, thiolate or activated aryloxy), arsenicals, sulfur and nitrogen mustard backbones, and cyanide-releasing motifs. Tier B covers phosgene-derived acyl halides and chloroformates, halogenated oximes, α-halo carbonyls, and groups making a compound impractical rather than agent-like (halosilanes, sulfonyl halides, peroxy acids). Tier C records a simple phosphoryl ester without a leaving group and is noted rather than excluded. An earlier version that did not distinguish leaving groups flagged DMMP at the same tier as sarin and admitted potassium cyanide to the faithful-and-safe quadrant; both were corrected.

2.8. Reproducibility

All analyses are deterministic given the seed and the permutation test is exhaustive rather than sampled. Numerical work used NumPy 2.5.3 [53], SciPy 1.18.1 [54], pandas 3.0.5 [55] and scikit-learn 1.9.1, figures Matplotlib 3.11.2 [56], and descriptors RDKit 2026.03.6 [31]. Code, descriptor tables, parsed toxicity values with their sources and the figure scripts are in the repository cited under Data and code availability.

3. Results

3.1. What the comparison contains decides the answer

On the thirty-five-compound panel, no authentic agent has a valid match to an incumbent simulant. Figure 2a shows why, and the mechanism is not the one the earlier version of this work reported. The distances from the agents to the incumbent simulants hardly move when the panel is extended: sarin sits at 1.14 typical nearest-neighbour separations from DMMP on the smaller panel and 1.19 on the larger, soman at 1.42 and 1.45, cyclosarin at 1.48 and 1.58. The compounds did not become less faithful. What changed is that diisopropyl fluorophosphate is present, at 1.07, 0.99 and 1.17 from the same three agents, and the match criterion requires a simulant to be nearer than every compound not designated one. The same holds across the panel, with 2-chloroethyl ethyl sulfide at 0.57 from sulfur mustard where dichloromethane stands at 1.67, and diethyl cyanophosphonate at 1.80 from tabun where DMMP stands at 2.58. Admitting the five as simulants gives five matched agents of eleven, against three on the twenty-eight-compound panel and zero on the primary one. The zero does not depend on where the threshold was set: no incumbent simulant is the nearest non-agent compound to any of the eleven agents, so the count stays at zero however far the threshold is relaxed.

That count is sensitive to the descriptors, and the honest statement of it is the interval rather than the point. Resampling the 41 merged-space descriptors with replacement (Section 2.6) gives the incumbent three a median of zero matched agents with a 95% interval of zero to three, and exactly zero in 58.8% of replicates, against a median of five and an interval of four to six for the eight compounds the field uses. The comparison is what survives: the difference between the counts is positive in 1000 of 1000 replicates, median four, interval one to six (Figure 2b). The per-agent reliabilities in the same replicates separate the firm results from the marginal ones. Sulfur mustard is matched by 2-chloroethyl ethyl sulfide in every replicate and tabun by diethyl cyanophosphonate in 52.5%, while under the incumbent set alone sarin recovers DMMP

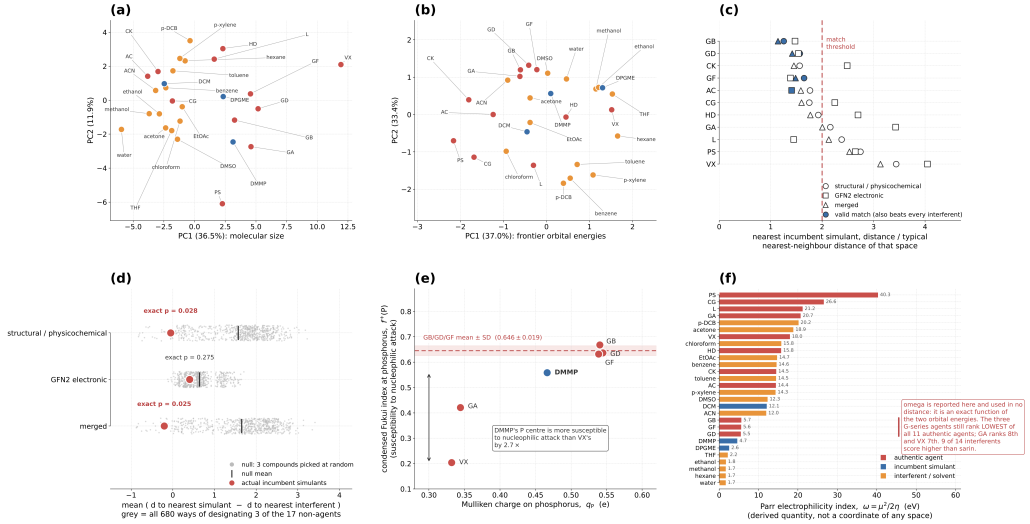


Figure 1: The twenty-eight-compound panel, which omits the five compounds the field also uses as simulants and is shown because the contrast with the primary panel of Figure 2 is this paper's result. (a) Principal components of the structural space and (b) of the three-coordinate GFN2-xTB electronic space. (c) Distance from each agent to its nearest incumbent simulant, normalised by the typical nearest-neighbour separation of the same space; filled markers are valid matches. (d) Exact permutation null over all 680 ways of designating three of the seventeen non-agents as simulants, with the actual designation marked. The p shown is the mean-difference statistic only; the other five and the adjusted values are in Table 1, where none reaches $q < 0.05$. (e) Local electrophilicity against Mulliken charge at phosphorus for the six phosphorus-bearing compounds (Supplementary Section S0.7). (f) The Parr electrophilicity index, an exact function of the orbital energies in panel (b), which enters no distance here (Section 2.3) and is shown because the three nerve agents rank lowest of the eleven agents on it.

as its nearest compound in 37.8% of replicates, soman in 14.4% and cyclosarin in 12.7%. The displacement of DMMP at the G agents is therefore the weakest part of the picture and the sulfur mustard result the strongest, which reverses what the point estimates alone would suggest.

Figure 1 gives the spaces. The leading structural component is a size axis carrying 36.5% of the variance and the leading electronic component loads on the two orbital energies against the dipole moment, the first two carrying 70.4%. Merging them returns a size axis at lower cumulative variance (45.9%) for a dimensional rather than a chemical reason, since at equal per-descriptor weighting thirty-eight structural descriptors outvote three electronic ones.

Table 1 gives every exact permutation test we can run across two panels, three spaces and six statistics, in full rather than selected from, because the pattern across the cells is the finding.

Three things hold across the table. On the primary panel all eighteen tests return $p > 0.15$: in no space and under no statistic are the incumbent simulants distinguishable from compounds drawn at random. Classing the five literature compounds as simulants recovers two nominal values below 0.05, both in spaces containing the structural descriptors, so the difficulty is not that the test cannot detect anything on this panel. On the twenty-eight-compound panel, which differs only in omitting those five and two organophosphorus pesticides, the incumbents beat random compounds under two of six statistics in the structural space and three of six in the merged space.

Correcting for the number of times we looked removes almost all of it, and the exception is instructive. Table 2 gives the five correction schemes of Section 2.4. On the primary panel the question does not arise, since the smallest nominal value is already 0.151. The schemes matter

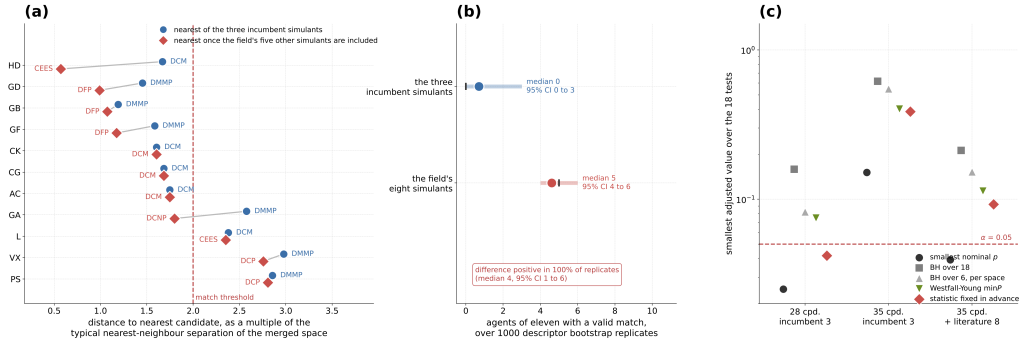


Figure 2: The composition of the comparison, not the fidelity of the compounds, is what the analysis turns on. (a) For each authentic agent, the distance to the nearest of the three incumbent simulants (circles) and to the nearest candidate once the five compounds the field also uses as simulants are included (diamonds), both as multiples of the typical nearest-neighbour separation of the merged space on the thirty-five-compound panel. A valid match requires a normalised distance below the dashed threshold *and* a distance smaller than that to every compound not designated a simulant, so sarin, soman and cyclosarin fail on the primary panel despite lying inside the threshold. (b) Matched agents under 1000 descriptor bootstrap replicates (Section 2.6) for the incumbent three and for the eight compounds the field uses; circles are means, vertical rules medians, bars 95% intervals. (c) Smallest adjusted value over the eighteen tests of each panel configuration under five ways of accounting for the search (Table 2).

Table 1: Every exact permutation test available. One-sided p for six distance statistics (Section 2.4) in three spaces, on two panels (Section 2.1). q is the smallest Benjamini-Hochberg value over the eighteen tests of that configuration. Bold marks nominal $p < 0.05$; no cell reaches $q < 0.05$.

Panel / simulant set	Space	Dim.	$\Delta\bar{d}$	$\Delta\tilde{d}$	$\overline{\ln r}$	$\bar{d}_{\text{sim}}$	n_{closer}	n_{valid}	min q
35 cpd., incumbent 3	structural	38	0.326	0.343	0.404	0.237	1.000	1.000	
(primary)	electronic	3	0.386	0.374	0.412	0.344	0.723	0.664	
	merged	41	0.283	0.151	0.366	0.183	1.000	1.000	0.617
35 cpd., incumbent	structural	38	0.062	0.341	**0.050**	0.158	0.319	0.159	
+ literature 8	electronic	3	0.121	0.086	0.123	0.127	0.165	0.080	
	merged	41	**0.053**	0.326	**0.039**	0.128	0.319	0.159	0.212
28 cpd., incumbent 3	structural	38	**0.028**	0.176	**0.044**	0.085	0.110	0.216	
(omits the five)	electronic	3	0.275	0.175	0.293	0.235	0.757	0.757	
	merged	41	**0.025**	0.124	**0.032**	**0.041**	0.110	0.235	0.159

Statistics, in order: mean and median of $d_{\text{sim}} - d_{\text{intf}}$; mean $\ln(d_{\text{sim}}/d_{\text{intf}})$; mean d_{sim}; count of agents with a simulant closer than any interferent; count of valid matches. Permutations: $\binom{17}{3} = 680$, $\binom{24}{3} = 2024$, $\binom{24}{8} = 735\,471$, enumerated exhaustively in each case.

only for the twenty-eight-compound panel. There the eighteen-test family gives a smallest adjusted value of 0.159, correcting the six statistics within each space gives 0.082, and the Westfall-Young min-P procedure, which uses the shared permutations and so accounts for the dependence between the tests exactly rather than approximating it, gives 0.075. That is the closest any correction brings the panel to the conventional threshold without crossing it. The Li-Ji approximation it replaces estimates 5.00 effective tests of eighteen and gives 0.125, so the approximation is conservative here by a factor near 1.7. Nothing in Table 1 reaches 0.05 on any panel under those four schemes.

The fifth scheme behaves differently. Had the mean difference been specified in advance as

the single statistic, three tests would have been run and the twenty-eight-compound structural and merged results would survive at $q = 0.042$, while neither thirty-five-compound configuration would (0.386 and 0.092). We did not specify a statistic in advance; we computed six and chose among them after seeing them, which is the circumstance the correction is for. The comparison is the paper's own argument made with its own data: one analysis, one data set, five defensible ways of accounting for the search, and a result significant under one of them and not the other four. The declaration that would have made the difference is one nobody in this literature makes, and it has to be made before the data are seen.

Table 2: Five ways of accounting for the search, applied to the eighteen tests of each panel configuration. The Westfall-Young min-P value [57] is exact for the dependence among the tests, since all eighteen share one enumeration of designations; m_{eff} approximates that dependence from the eigenvalues of the correlation matrix of the eighteen nulls [58], used as a Bonferroni divisor. Only the pre-specified-statistic scheme, on the smaller panel, returns anything below 0.05.

Accounting scheme	35 cpd., incumbent 3 (primary)	35 cpd., + literature 8	28 cpd., incumbent 3 (omits the five)
Smallest nominal p (18 tests)	0.151	0.039	0.025
Benjamini-Hochberg over 18	0.617	0.212	0.159
Benjamini-Hochberg over 6, per space	0.548	0.153	0.082
Westfall-Young min-P over 18	0.402	0.113	0.075
m_{eff} of 18 (Li-Ji)	6.00	5.00	5.00
Bonferroni at m_{eff}	0.907	0.196	0.125
Statistic specified in advance (3 tests)	0.386	0.092	**0.042**

Two further observations bound what the nominal values can mean. The count of agents for which a compound outside the designated set is nearer than any incumbent simulant is 11 of 11 in the merged space of the primary panel against a null expectation of 9.63 set by the 21:3 group-size asymmetry alone, and 6, 10 and 6 of 11 in the three spaces of the smaller panel against 9.06, so the raw count carries no evidence without the permutation test and in one space runs the wrong way. On the thirty-five-compound panel with eight simulants, leaving out one agent at a time moves the structural p between 0.017 and 0.170, the worst case being removal of tabun; a result that one of eleven agents can carry across the 0.05 line is not a property of the panel.

3.2. The difference between the spaces is not dimensional, and the nominal value is not informative

The structural space has 38 coordinates and the electronic space has 3, so a difference between them could be geometric rather than chemical. It is not. Truncating the standardised structural space to its first three principal components strengthens the result rather than weakening it, from $p = 0.028$ to $p = 0.0044$ on the twenty-eight-compound panel, which leaves the two spaces 62-fold apart in p at matched dimension. The second control is the one that matters. Of 2000 random three-descriptor subsets of the structural space, 36.3% reach $p < 0.05$, and the observed 38-dimensional value of 0.028 sits at the 25.8th percentile of that distribution while the electronic value of 0.275 sits at the 77.9th. Both are ordinary draws. The rate itself is inflated by correlation among the descriptors, falling to 16.9% on whitened coordinates, but the argument rests on the percentile and that is unaffected. Both controls are reported in full in Supplementary Section S0.6. The spaces do differ, and the difference is chemical rather than geometric; what the data do not support is that the difference says anything about the compounds that were designated.

3.3. How many agents have a valid simulant depends on the panel

Table 3 lists, for each agent, the nearest incumbent simulant in the merged space as a multiple of that space's typical nearest-neighbour separation. Sarin (1.13), soman (1.40) and cyclosarin (1.47) are matched, all three by DMMP; tabun at 2.03 and VX at 3.13 are not. DPGME is nominally VX's nearest simulant in all three spaces, but at 3.13 typical separations in the merged space and 3.44 in the structural space that is not a match in any useful sense. Dichloromethane, used as a pulmonary-agent simulant, is beaten for phosgene by acetone (3.91) and ethyl acetate (3.94) against its own 6.02 in the structural and merged spaces, and leads ethyl acetate only in the electronic space (2.18 against 2.28).

Table 3: Nearest incumbent simulant for each agent, as a multiple of the typical nearest-neighbour distance of the merged space. A match requires a normalised distance ≤ 2 and a distance smaller than that to every interferent.

Authentic agent	Nearest simulant	Normalised distance	Valid match
GB	DMMP	1.13	yes
GD	DMMP	1.40	yes
CK	DCM	1.44	no
GF	DMMP	1.47	yes
AC	DCM	1.57	no
CG	DCM	1.64	no
HD	DCM	1.75	no
GA	DMMP	2.03	no
L	DCM	2.12	no
PS	DMMP	2.70	no
VX	DPGME	3.13	no

The count is stable but not invariant. At thresholds of 1.75 and 2.0 the merged space returns {GB, GD, GF} and at 2.5 and 3.0 it returns four, tabun being admitted once the bar passes its value of 2.03; the structural space behaves identically, except that at 1.5 it narrows to {GB} alone as soman (1.567) and cyclosarin (1.648) fall outside. The three-agent figure is therefore a statement about thresholds of 1.75 and above. The electronic space returns four matches at every threshold tested, the extra one being hydrogen cyanide at 1.335 matched by dichloromethane, and in no space at any threshold does the count exceed four of eleven. The headline figure of three refers to the merged space tabulated here, and the qualification is not cosmetic: the agent the electronic space adds is matched by dichloromethane rather than by DMMP.

The dependence on the panel is sharper than the dependence on the threshold. On the thirty-five-compound panel, with the incumbent three still the only simulants and the five literature compounds in the comparison pool, the merged space returns no valid match for any of the eleven agents: DMMP loses GB, GD and GF to diisopropyl fluorophosphate, which is nearer to all three. Admitting the five as simulants gives five matches of eleven, more than the incumbent set achieves on either panel, with DFP taking sarin, soman and cyclosarin at 1.07, 0.99 and 1.17, diethyl cyanophosphonate taking tabun at 1.80, and 2-chloroethyl ethyl sulfide taking sulfur mustard at 0.57, the closest agent-simulant pair anywhere in this work. VX remains unmatched in every panel and space we examined.

We draw no conclusion from this that the field should switch compounds, because the five were not selected by any procedure we can inspect either, and Section 3.1 applies to them exactly as it applies to the incumbent three. What the comparison establishes is narrower: the statement

"three of eleven agents have a valid simulant" is true of one panel, false of a second, and replaced by a different list on a third, which is the same instability the permutation tests report and not an independent finding.

3.4. What the array actually measures

Every argument above rests on descriptor distances, and the premise that one of those spaces corresponds to what a vapour-phase sensor reports is tested in Supplementary Section S0.8 against measured response from a 29-dye colorimetric array covering twelve of the panel compounds. The test does not resolve the question in either direction. Of eight panel-channel-space cells, two reach a nominal $p < 0.05$ under Mantel permutation over compounds, and they favour different descriptor spaces; corrected across the eight, the smallest adjusted value is 0.098 and no cell reaches $q < 0.05$. With six compounds per plate set the design detects $|\rho| \geq 0.669$ at 80% power, so these data exclude a strong association between either descriptor space and the measured response and nothing weaker. Two artifacts of array analysis of this kind are reported there because our own first version contained both: dye-axis centring fixes the between-compound mean cosine at $-1/(n-1)$, which the observed means clear by only 0.013 to 0.066, so a negative between-compound similarity is arithmetic rather than evidence; and the measured agent nearest to DMMP in the ultraviolet channel is tabun rather than sarin, an agent the descriptor criterion of Section 3.3 excludes. The claims in the rest of this paper are written to that limit.

3.5. The toxicity half of the criterion

The criterion has a second half: that compounds chosen on physicochemical grounds present minimal toxicity risk [16]. We tested it on the same panel and report the analysis in full in the Supplementary Material, since its conclusions stand on their own data. Three results are needed here.

Dichloromethane has a regulator-derived acute inhalation value and DMMP has none, EPA having declined to derive an inhalation reference concentration for want of data (Supplementary Section S0.1). The two incumbents therefore differ in how well documented they are and not only in how hazardous, which matters because DMMP is also the compound against which decontamination efficiency is quantified [15]. Dichloromethane's value is favourable on the acute axis, at the 96th percentile of a 207-compound pool, and that ranking says nothing about the chronic and carcinogenic classification now displacing it from solvent inventories.

A model cannot fill the gap. A random forest trained on 230 AEGL chemicals reaches $R^2 = 0.458$ and then predicts every one of twenty held-out military agents safer than measured, by factors of 2.5 to 2600, while certifying fourteen of them as inside its applicability domain (Supplementary Section S0.2). The failure is not a missing mechanism, since thirteen training compounds carry an acetylcholinesterase-inhibiting pharmacophore and one is more acutely toxic than sarin. It is tail regression, a bias defined in the response space that an input-space domain check cannot see.

Fidelity and acute toxicity are independent across the 207 compounds carrying both measured axes (Spearman $r = +0.004$, 95% CI -0.132 to $+0.143$), so the two can be improved separately rather than traded against each other (Supplementary Section S0.11). We draw no candidate recommendation from that: one resting on a single panel and two axes would repeat the error this paper reports.

4. Discussion

The criterion by which low-toxicity simulants are chosen is stated clearly enough to test [16]: align hydrophobicity, volatility and solubility with the agent. What the practice does not state is what the chosen compounds should be compared against, and that decides the answer. On a panel containing the compounds the field itself uses as simulants, the three incumbents match no agent and no test falls below $p = 0.15$; on a panel omitting them, three agents are matched and the smallest nominal value is 0.025.

The mechanism is not a story about fidelity. The incumbents sit almost as close to the agents on one panel as on the other, moving by 0.03 to 0.10 typical separations (Section 3.1). What the larger panel supplies is diisopropyl fluorophosphate, nearer to sarin, soman and cyclosarin than DMMP is, so a criterion that asks a simulant to beat the alternatives returns nothing. Resampling the descriptors leaves that comparison intact in every replicate while showing the individual displacements to differ in firmness: sulfur mustard is matched by 2-chloroethyl ethyl sulfide in all 1000 replicates, and sarin recovers DMMP as its nearest compound in 37.8%. A designation defended by a comparison that excludes the alternatives is not defended.

An earlier version of this work reported the twenty-eight-compound structural cell, $p = 0.028$, as the finding and built a thesis on the contrast between it and the electronic space. Three things undo that reading, and each is a mistake another group could make the same way. The value does not survive the eighteen tests of its own family (Benjamini-Hochberg $q = 0.159$), although it would have survived at $q = 0.042$ had the statistic been fixed in advance, which demonstrates the paper's argument rather than defending the value. It sits at the 25.8th percentile of 2000 arbitrary three-descriptor slices of the same block. And it is specific to the panel that omits the five.

Two things we expected to dissolve the contrast between the spaces did not. One of the four electronic descriptors was the Parr electrophilicity index, an exact function of two of the others, so the electronic space never had the dimension we reported; removing it moved the electronic p only from 0.297 to 0.275, so the descriptor was misdescribed rather than manufacturing the contrast. Matching the dimensions did not dissolve it either (Section 3.2). The contrast is real; what fails is the inference from it, because the structural value anchoring it is an ordinary draw from the subspaces we could have chosen.

What remains is a claim about comparison rather than about chemistry. On the primary panel we cannot establish that the incumbent simulants are closer to the agents than compounds drawn at random, in any space, under any of six statistics, under any of five ways of accounting for the search, and we cannot establish the opposite either. What we can establish is that the compounds which would make the comparison informative were not in it.

The documentary record explains how that arose, and it is not a story about carelessness. Two of the three compounds were designated, and designated carefully, for measurements that are not the one they are now asked to support (Section 2.1): DMMP for adsorptive breakthrough on a filter bed [27], DPGME for ion-mobility detectability in a training composition carrying a fluorescent dye [28]. Each selection was fit for the measurement it was made for. The compounds then carried their status into sensor development, where the governing property is the response of a dye or an electrode rather than a breakthrough time or an ion mobility, and nothing in the original validation speaks to that. Dichloromethane did not make that journey; we found no designating document for it at all.

That mechanism is general. A simulant is validated against a measurement, not against an agent, and treating the validation as a property of the compound, so that it travels into a new application, produces a working set whose standing no descriptor space can reproduce. The

corrective is not a replacement compound. It is to state at the point of use which measurement a simulant was validated against, and to fix the panel, the space and the statistic before testing a claim that it is valid for a different one. Agrawal et al. [17] reach the same place from the other direction: adequacy is a property of the compound together with the measurement it stands in for, and neither is recoverable from the compound alone.

Three consequences follow for practice. DMMP is a surrogate for one G agent at a time rather than for the G series, and not a V-agent surrogate at all: the descriptor criterion assigns GB, GD and GF to it, while the measured array puts sarin first in reflectance and tabun, which fails that criterion, first in the ultraviolet channel (Supplementary Section S0.8), and a sensor calibrated on DMMP meets a phosphorus centre 2.7 times more electrophilic than VX presents (Supplementary Section S0.7). The global electrophilicity index should not enter a descriptor space, for an arithmetic rather than a chemical reason: where the frontier orbital energies are present, ω adds no information, and a correlation filter will not notice, because exact functional dependence is compatible with a correlation of -0.899. And the safety of a simulant is a question about evidence before it is one about hazard, since dichloromethane carries a regulator-derived acute inhalation value and DMMP carries none (Section 3.5).

We did not attempt to improve the toxicity model by training on organophosphorus compounds, and the distinction is narrower than a refusal to model the chemistry. Mechanistic structure–activity models of organophosphorus lethality exist and we have contributed to them [26]; such a model is an instrument for assessing compounds that already exist. What we decline to build is the same instrument turned into a design objective, a predictor of mammalian lethality evaluated over structures not yet made inside a loop that proposes them, which is the configuration in which a safety model becomes a potency model [24]. The cost is an acknowledged gap: for the compound class of greatest interest we can say that a predicted value is unreliable and cannot supply a better one. Reporting the gap rather than closing it is the correct outcome, and where to set the boundary between publishing a capability and withholding it is contested rather than settled [25].

Limitations.. The documentary findings of Section 2.1 rest on public sources. National CBRN defence manuals are generally not public, and a designation in one of them would be invisible to the search we ran; the same applies to two Test Operations Procedures we identified by title but could not obtain. The claim is therefore that no public instrument designates these three compounds for the use the sensor literature makes of them, not that no instrument does. The agent set is eleven compounds and the nearest-neighbour statistics inherit that: on the larger panel, leaving out one agent at a time moves the structural p between 0.017 and 0.170. The permutation test is directional and reported one-sided, with both two-sided conventions given in Section 2.4. The two panels are the two we could build from compounds carrying both descriptor and toxicity data, and a third is possible, since we class dichlorvos and malathion as interferents although both carry the phosphoryl motif the structural filter treats as agent-class; a reader who classed them as simulants would obtain numbers we do not report, so the three configurations of Table 1 sample the configuration space rather than census it. The measured-array analysis rests on six or seven compounds per plate set at four concentrations, so its intervals describe variation across concentrations within one set rather than agreement between preparations (Supplementary Section S0.8). GFN2-xTB is semi-empirical, and the quantities used here are relative within a consistent protocol rather than absolute.

5. Conclusions

Simulant validity is not a property of a compound. It is a property of a compound set against a comparison, and the practice of choosing chemical warfare agent simulants does not record what the comparison should contain. On a panel of thirty-five compounds that includes the five the sensor, sorbent and decontamination literature also uses as simulants, not one of eleven authentic agents has a valid match to an incumbent simulant and not one of eighteen tests falls below $p = 0.15$. The incumbents are no further from the agents on that panel than on the smaller one; diisopropyl fluorophosphate is simply nearer, and under descriptor resampling the incumbent three match a median of zero agents against five for the compounds the field uses, the difference positive in every one of 1000 replicates. Omitting those five restores three matches and a nominal structural p of 0.028, a value at the 25.8th percentile of 2000 arbitrary three-descriptor subspaces of the same block that survives one of five defensible ways of accounting for the search.

The documentary record gives the reason. DMMP was designated for filtration breakthrough testing and DPGME for ion-mobility training compositions, each by a specific instrument and for a measurement that is not detection chemistry; for dichloromethane we found no designating instrument at all. A simulant is validated against a measurement rather than against an agent, and that validation does not travel with the compound into a new application.

The absence of evidence here is not evidence of unsuitability, and we draw no conclusion that the incumbent simulants are poor choices. What we conclude is narrower and addressed to method. A simulant claim is testable, this field has the data to test it, and the test returns whatever the analyst's three undeclared choices imply. The practical form of that conclusion is a reporting rule: a simulant result should name the measurement the compound was validated against and the comparison set it was tested in, and should fix the panel, the descriptor space and the statistic before the test is run. Until those are declared in advance, a statement that a compound is a validated simulant reports a convention rather than a measurement.

Findings from the toxicity axis are in the Supplementary Material and do not depend on the contested comparison. DMMP has no regulator-derived acute inhalation value, the agency having declined to derive one, while dichloromethane has one favourable on the acute axis alone, so the two incumbents differ in how well documented they are and not only in how hazardous. A random forest trained on 230 AEGL chemicals predicts all twenty held-out military agents safer than measured, by factors of 2.5 to 2600, so predicted toxicity cannot stand in for the missing values (Supplementary Section S0.10). Fidelity and acute toxicity are independent across the 207 compounds carrying both axes, and eight compounds combine top-quartile fidelity with measured safety within one order of magnitude of dichloromethane (Supplementary Section S0.11). We propose none as a replacement, because proposing one would repeat the error this paper is about.

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CRediT authorship contribution statement

Ku Kang: conceptualisation, methodology, software, formal analysis, investigation, data curation, visualisation, writing – original draft. **Yoon Jeong Jang:** validation, writing – review and editing, supervision, project administration.

Declaration of competing interest

The authors declare no competing financial interest.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used a large language model (Anthropic Claude) to assist in drafting and editing prose, in writing and auditing the analysis and figure scripts, and in building the internal-consistency checks described under Reproducibility. No numerical result was produced by the model: every value reported here is read from the output of the deposited scripts, which were executed to generate it. The authors reviewed and edited all content and take full responsibility for it.

Data and code availability

Descriptor tables, parsed AEGL values with their sources, permutation distributions, candidate tables and all analysis and figure scripts are in a public repository at `https://github.com/bisu9082/simulant-designation-evidence`, at the tagged release `v1.0`, and are also provided with this article. The repository also carries the primary EPA source document from which every toxicity value here was parsed, so the parse can be checked against the original. The exact permutation test is exhaustive and the remaining analyses are deterministic given the stated seed.

References

[1] K. Kim, O. G. Tsay, D. A. Atwood, D. G. Churchill, Destruction and detection of chemical warfare agents, Chemical Reviews 111 (9) (2011) 5345–5403. doi:10.1021/cr100193y.

[2] Y. J. Jang, K. Kim, O. G. Tsay, D. A. Atwood, D. G. Churchill, Update 1 of: Destruction and detection of chemical warfare agents, Chemical Reviews 115 (24) (2015) PR1–PR76. doi:10.1021/acs.chemrev.5b00402.

[3] S. Royo, R. Martínez-Máñez, F. Sancenón, A. M. Costero, M. Parra, S. Gil, Chromogenic and fluorogenic reagents for chemical warfare nerve agents' detection, Chemical Communications (46) (2007) 4839. doi:10.1039/b707063b.

[4] M. R. Sambrook, S. Notman, Supramolecular chemistry and chemical warfare agents: from fundamentals of recognition to catalysis and sensing, Chemical Society Reviews 42 (24) (2013) 9251. doi:10.1039/c3cs60230c.

[5] L. M. Eubanks, T. J. Dickerson, K. D. Janda, Technological advancements for the detection of and protection against biological and chemical warfare agents, Chemical Society Reviews 36 (3) (2007) 458. doi:10.1039/b615227a.

[6] N. A. Rakow, K. S. Suslick, A colorimetric sensor array for odour visualization, Nature 406 (6797) (2000) 710–713. doi:10.1038/35021028.

[7] H. Lin, M. Jang, K. S. Suslick, Preoxidation for colorimetric sensor array detection of VOCs, Journal of the American Chemical Society 133 (42) (2011) 16786–16789. doi:10.1021/ja207718t.

[8] J. R. Askim, M. Mahmoudi, K. S. Suslick, Optical sensor arrays for chemical sensing: the optoelectronic nose, Chemical Society Reviews 42 (22) (2013) 8649. doi:10.1039/c3cs60179j.

[9] J. Kim, K. Kang, M. Shin, S. Kim, J. Yoo, M.-K. Kim, W. B. Lee, D.-H. Lee, Machine learning-enhanced colorimetric sensor array for rapid detection of nerve agents, Journal of Hazardous Materials 501 (2026) 140703. doi:10.1016/j.jhazmat.2025.140703.

[10] S. Bae, K. Kang, Y. K. Kim, Y. J. Jang, D.-H. Lee, Field-deployable real-time AI system for chemical warfare agent detection using YOLOv8 and colorimetric sensors, Chemometrics and Intelligent Laboratory Systems 261 (2025) 105365. doi:10.1016/j.chemolab.2025.105365.

[11] M. Shin, K. Kang, J. Kim, S. Kim, J. Yoo, Y. J. Jang, M.-K. Kim, E. V. Anslyn, D.-H. Lee, Portable machine learning–integrated optical sensor array for rapid and selective detection of authentic Novichok nerve agents, Sensors and Actuators B: Chemical 457 (2026) 139706. doi:10.1016/j.snb.2026.139706.

[12] S. L. Bartelt-Hunt, D. R. U. Knappe, M. A. Barlaz, A review of chemical warfare agent simulants for the study of environmental behavior, Critical Reviews in Environmental Science and Technology 38 (2) (2008) 112–136. doi:10.1080/10643380701643650.

[13] J. E. Mondloch, M. J. Katz, W. C. I. III, P. Ghosh, P. Liao, W. Bury, G. W. Wagner, M. G. Hall, J. B. DeCoste, G. W. Peterson, R. Q. Snurr, C. J. Cramer, J. T. Hupp, O. K. Farha, Destruction of chemical warfare agents using metal–organic frameworks, Nature Materials 14 (5) (2015) 512–516. doi:10.1038/nmat4238.

[14] S.-Y. Moon, Y. Liu, J. T. Hupp, O. K. Farha, Instantaneous hydrolysis of nerve-agent simulants with a six-connected zirconium-based metal–organic framework, Angewandte Chemie International Edition 54 (23) (2015) 6795–6799. doi:10.1002/anie.201502155.

[15] H. S. Chi, J.-m. Park, J. Yoo, G. J. Choi, S. H. Hong, J. Son, J.-G. Lee, K. Kang, Feasibility study of electron beam irradiation for decontamination of chemical and biological warfare agent simulants: Dosimetric and operational evaluation, Applied Radiation and Isotopes 232 (2026) 112582. doi:10.1016/j.apradiso.2026.112582.

[16] S. Webb, F. Coulon, T. Temple, A critical review of liquid, low toxicity chemical warfare agent simulants: Enhancing accuracy, safety, and methodological approaches for sampling, Journal of Hazardous Materials 492 (2025) 138021. doi:10.1016/j.jhazmat.2025.138021.

[17] M. Agrawal, D. F. Sava Gallis, J. A. Greathouse, D. S. Sholl, How useful are common simulants of chemical warfare agents at predicting adsorption behavior?, The Journal of Physical Chemistry C 122 (45) (2018) 26061–26069. doi:10.1021/acs.jpcc.8b08856.

[18] H. S. Chi, E. Kim, J. Kim, K.-D. Byun, K. Kang, Chemical protective suit performance against TICs and CWAs: permeation resistance and PPE safety evaluation, Next Research 2 (3) (2025) 100688. doi:10.1016/j.nexres.2025.100688.

[19] S. Grimme, C. Bannwarth, P. Shushkov, A robust and accurate tight-binding quantum chemical method for structures, vibrational frequencies, and noncovalent interactions of large molecular systems parametrized for all spd-block elements (z = 1–86), Journal of Chemical Theory and Computation 13 (5) (2017) 1989–2009. doi:10.1021/acs.jctc.7b00118.

[20] C. Bannwarth, S. Ehlert, S. Grimme, GFN2-xTB—an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions, Journal of Chemical Theory and Computation 15 (3) (2019) 1652–1671. doi:10.1021/acs.jctc.8b01176.

[21] C. Bannwarth, E. Caldeweyher, S. Ehlert, A. Hansen, P. Pracht, J. Seibert, S. Spicher, S. Grimme, Extended tight-binding quantum chemistry methods, WIREs Computational Molecular Science 11 (2) (2020) e1493. doi:10.1002/wcms.1493.

[22] A. Watson, D. Opresko, R. Young, V. Hauschild, Development and application of acute exposure guideline levels (AEGLs) for chemical warfare nerve and sulfur mustard agents, Journal of Toxicology and Environmental Health, Part B 9 (3) (2006) 173–263. doi:10.1080/15287390500194441.

[23] K. Kang, D.-H. Lee, J. Kim, J. Yoo, Condition-dependent risk cartography of brominated disinfection by-products in seawater desalination, Journal of Environmental Chemical Engineering 14 (5) (2026) 125179. doi:10.1016/j.jece.2026.125179.

[24] F. Urbina, F. Lentzos, C. Invernizzi, S. Ekins, Dual use of artificial-intelligence-powered drug discovery, Nature Machine Intelligence 4 (3) (2022) 189–191. doi:10.1038/s42256-022-00465-9.

[25] D. Kim, J. Yoo, K. Kang, Exposure and disclosure in countering CBRN threats: implications for nonproliferation strategy, The Nonproliferation Review 33 (4-6) (2026) 283–306. doi:10.1080/10736700.2026.2685340.

[26] Y. Jang, J. Kim, D.-H. Lee, J. Yoo, J. Park, K. Kang, Hybrid QSAR modeling of LD50 in organophosphorus nerve agents: a mechanistic approach using DFT and molecular docking, Computational Toxicology 35 (2025) 100363. doi:10.1016/j.comtox.2025.100363.

[27] National Institute for Occupational Safety and Health, Chemical warfare agent (CWA) simulant program for respirator materials resistance testing, Tech. rep., National Personal Protective Technology Laboratory, NIOSH, dated 31 July 2002 (Jul. 2002).
URL https://archive.cdc.gov/www_cdc_gov/niosh/npptl/standardsdev/cbrn/apr/concepts/pdfs/simulantjul31-508.pdf

[28] A. T. Seitzinger, J. A. Genovese, Chemical agent simulant training composition, U.S. Patent US 7,129,094 B1, assigned to the United States Department of the Army. Filed 16 May 2003; granted 31 October 2006 (2006).
URL https://patents.google.com/patent/US7129094B1/en

[29] National Research Council, Technical Assessment of the Man-in-Simulant Test Program, The National Academies Press, Washington, DC, 1997. doi:10.17226/5936.

[30] D. S. Lawrence, K. A. Van Houten, Simulants for chemical and explosive threats, Johns Hopkins APL Technical Digest 37 (4) (2025).

[31] G. Landrum, et al., RDKit: Open-source cheminformatics, https://www.rdkit.org, version 2026.03.6. Cited by name, version and project URL: the Zenodo DOIs registered for RDKit identify individual releases (10.5281/zenodo.10398 is release 2014.03.1) and no version-independent concept DOI could be verified, so none is given here (2026).

[32] S. A. Wildman, G. M. Crippen, Prediction of physicochemical parameters by atomic contributions, Journal of Chemical Information and Computer Sciences 39 (5) (1999) 868–873. doi:10.1021/ci9903071.

[33] P. Ertl, B. Rohde, P. Selzer, Fast calculation of molecular polar surface area as a sum of fragment-based contributions and its application to the prediction of drug transport properties, Journal of Medicinal Chemistry 43 (20) (2000) 3714–3717. doi:10.1021/jm000942e.

[34] L. B. Kier, Shape indexes of orders one and three from molecular graphs, Quantitative Structure-Activity Relationships 5 (1) (1986) 1–7. doi:10.1002/qsar.19860050102.

[35] D. Rogers, M. Hahn, Extended-connectivity fingerprints, Journal of Chemical Information and Modeling 50 (5) (2010) 742–754. doi:10.1021/ci100050t.

[36] S. Kim, J. Chen, T. Cheng, A. Gindulyte, J. He, S. He, Q. Li, B. A. Shoemaker, P. A. Thiessen, B. Yu, L. Zaslavsky, J. Zhang, E. E. Bolton, PubChem 2025 update, Nucleic Acids Research 53 (D1) (2024) D1516–D1525. doi:10.1093/nar/gkae1059.

[37] A. J. Williams, C. M. Grulke, J. Edwards, A. D. McEachran, K. Mansouri, N. C. Baker, G. Patlewicz, I. Shah, J. F. Wambaugh, R. S. Judson, A. M. Richard, The CompTox chemistry dashboard: a community data resource for environmental chemistry, Journal of Cheminformatics 9 (1) (2017) 61. doi:10.1186/s13321-017-0247-6.

[38] S. Ehlert, M. Stahn, S. Spicher, S. Grimme, Robust and efficient implicit solvation model for fast semiempirical methods, Journal of Chemical Theory and Computation 17 (7) (2021) 4250–4261. doi:10.1021/acs.jctc.1c00471.

[39] S. Wang, J. Witek, G. A. Landrum, S. Riniker, Improving conformer generation for small rings and macrocycles based on distance geometry and experimental torsional-angle preferences, Journal of Chemical Information and Modeling 60 (4) (2020) 2044–2058. doi:10.1021/acs.jcim.0c00025.

[40] S. Riniker, G. A. Landrum, Better informed distance geometry: Using what we know to improve conformation generation, Journal of Chemical Information and Modeling 55 (12) (2015) 2562–2574. doi:10.1021/acs.jcim.5b00654.

[41] T. A. Halgren, Merck molecular force field. I. basis, form, scope, parameterization, and performance of MMFF94, Journal of Computational Chemistry 17 (5-6) (1996) 490–519. doi:10.1002/(sici)1096-987x(199604)17:5/6<490::aid-jcc1>3.0.co;2-p.

[42] A. K. Rappe, C. J. Casewit, K. S. Colwell, W. A. Goddard, W. M. Skiff, UFF, a full periodic table force field for molecular mechanics and molecular dynamics simulations, Journal of the American Chemical Society 114 (25) (1992) 10024–10035. doi:10.1021/ja00051a040.

[43] P. Pracht, F. Bohle, S. Grimme, Automated exploration of the low-energy chemical space with fast quantum chemical methods, Physical Chemistry Chemical Physics 22 (14) (2020) 7169–7192. doi:10.1039/c9cp06869d.

[44] R. G. Parr, L. v. Szentpály, S. Liu, Electrophilicity index, Journal of the American Chemical Society 121 (9) (1999) 1922–1924. doi:10.1021/ja983494x.

[45] K. Wiberg, Application of the Pople–Santry–Segal CNDO method to the cyclopropylcarbinyl and cyclobutyl cation and to bicyclobutane, Tetrahedron 24 (3) (1968) 1083–1096. doi:10.1016/0040-4020(68)88057-3.

[46] I. Mayer, Charge, bond order and valence in the *ab initio* SCF theory, Chemical Physics Letters 97 (3) (1983) 270–274. doi:10.1016/0009-2614(83)80005-0.

[47] R. G. Parr, W. Yang, Density functional approach to the frontier-electron theory of chemical reactivity, Journal of the American Chemical Society 106 (14) (1984) 4049–4050. doi:10.1021/ja00326a036.

[48] W. Yang, W. J. Mortier, The use of global and local molecular parameters for the analysis of the gas-phase basicity of amines, Journal of the American Chemical Society 108 (19) (1986) 5708–5711. doi:10.1021/ja00279a008.

[49] P. Geerlings, F. D. Proft, W. Langenaeker, Conceptual density functional theory, Chemical Reviews 103 (5) (2003) 1793–1874. doi:10.1021/cr990029p.

[50] L. Domingo, M. Ríos-Gutiérrez, P. Pérez, Applications of the conceptual density functional theory indices to organic chemistry reactivity, Molecules 21 (6) (2016) 748. doi:10.3390/molecules21060748.

[51] R. BENIGNI, C. BOSSA, Structure alerts for carcinogenicity, and the *Salmonella* assay system: A novel insight through the chemical relational databases technology, Mutation Research/Reviews in Mutation Research 659 (3) (2008) 248–261. doi:10.1016/j.mrrev.2008.05.003.

[52] S. J. Enoch, C. M. Ellison, T. W. Schultz, M. T. D. Cronin, A review of the electrophilic reaction chemistry involved in covalent protein binding relevant to toxicity, Critical Reviews in Toxicology 41 (9) (2011) 783–802. doi:10.3109/10408444.2011.598141.

[53] C. R. Harris, K. J. Millman, S. J. v. d. Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. v. Kerkwijk, M. Brett, A. Haldane, J. F. d. Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, T. E. Oliphant, Array programming with numpy, Nature 585 (7825) (2020) 357–362. doi:10.1038/s41586-020-2649-2.

[54] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. v. d. Walt, M. Brett, J. Wilson,

K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. v. Mulbregt, A. Vijaykumar, A. P. Bardelli, A. Rothberg, A. Hilboll, A. Kloeckner, A. Scopatz, A. Lee, A. Rokem, C. N. Woods, C. Fulton, C. Masson, C. Häggström, C. Fitzgerald, D. A. Nicholson, D. R. Hagen, D. V. Pasechnik, E. Olivetti, E. Martin, E. Wieser, F. Silva, F. Lenders, F. Wilhelm, G. Young, G. A. Price, G.-L. Ingold, G. E. Allen, G. R. Lee, H. Audren, I. Probst, J. P. Dietrich, J. Silterra, J. T. Webber, J. Slavič, J. Nothman, J. Buchner, J. Kulick, J. L. Schönberger, J. V. d. M. Cardoso, J. Reimer, J. Harrington, J. L. C. Rodríguez, J. Nunez-Iglesias, J. Kuczynski, K. Tritz, M. Thoma, M. Newville, M. Kümmerer, M. Bolingbroke, M. Tartre, M. Pak, N. J. Smith, N. Nowaczyk, N. Shebanov, O. Pavlyk, P. A. Brodtkorb, P. Lee, R. T. McGibbon, R. Feldbauer, S. Lewis, S. Tygier, S. Sievert, S. Vigna, S. Peterson, S. More, T. Pudlik, T. Oshima, T. J. Pingel, T. P. Robitaille, T. Spura, T. R. Jones, T. Cera, T. Leslie, T. Zito, T. Krauss, U. Upadhyay, Y. O. Halchenko, Y. Vázquez-Baeza, Scipy 1.0: fundamental algorithms for scientific computing in python, Nature Methods 17 (3) (2020) 261–272. doi:10.1038/s41592-019-0686-2.

[55] W. McKinney, Data structures for statistical computing in python, 2010, pp. 56–61. doi:10.25080/majora-92bf1922-00a.

[56] J. D. Hunter, Matplotlib: A 2d graphics environment, Computing in Science & Engineering 9 (3) (2007) 90–95. doi:10.1109/mcse.2007.55.

[57] P. H. Westfall, S. S. Young, Resampling-Based Multiple Testing: Examples and Methods for *p*-Value Adjustment, Wiley, New York, 1993.

[58] J. Li, L. Ji, Adjusting multiple testing in multilocus analyses using the eigenvalues of a correlation matrix, Heredity 95 (3) (2005) 221–227. doi:10.1038/sj.hdy.6800717.